

SHAHZAD KHICHI

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PROFESSIONAL SUMMARY

Software Engineering student at FAST NUCES with strong expertise in full stack and backend development using MERN Stack, Next.js, NestJS, FastAPI, Spring Boot, and TypeScript. Experienced in building scalable real-time applications, RESTful APIs, and enterprise-style systems with a strong foundation in OOP, DSA, Low-Level Design, and software architecture. Passionate about backend engineering, clean code practices, AI-driven systems, and scalable software systems and backend technologies.

EDUCATION

FAST – National University of Computer and Emerging Sciences (FAST NUCES)

2023 – Present

BS Software Engineering — Coursework: DSA, OOP, Operating Systems, Database Systems, Software Design

EXPERIENCE

Freelance Full Stack Developer

2024 – Present

- Designed and developed scalable full-stack web applications using React, Next.js, Node.js, Express.js, and TypeScript; implemented 20+ RESTful API endpoints with JWT authentication, modular architecture, and clean separation of concerns.
- Integrated backend services with MongoDB and PostgreSQL; worked with deployment workflows, environment configurations, and production-ready application setups for scalable systems.
- Collaborated directly with clients to gather requirements, debug production issues, and continuously improve application performance and usability.

PROJECTS

Apex Health — Node.js · TypeScript · Prisma ORM · MongoDB · Redis · BullMQ · React · Socket.IO · WebRTC · Stripe

- Production-grade telehealth ecosystem — telemedicine, AI-assisted diagnostics, lab workflows, digital prescriptions, and scheduling — built on N-Tier architecture with Repository Pattern & Service Layer.
- JWT auth with refresh token rotation, RBAC, rate limiting, Socket.IO real-time chat, WebRTC video, Stripe payments, Google Gemini AI, and BullMQ async job processing.

APMS — Arabic Prose Management System — Java 17 · JavaFX · Maven · MySQL · JDBC · JUnit 5 · Mockito · NLP

- Enterprise-grade desktop application for Arabic prose management and NLP using Three-Tier Architecture with Facade, Strategy, State, Singleton, Observer, Factory Method, and Abstract Factory design patterns.
- Built an Arabic NLP pipeline (normalization, tokenization, stemming, root analysis, morphological processing via AlKhalil Morphological System 2) and multi-strategy search engine using Rabin-Karp rolling hash and Regex pattern matching.
- Designed a normalized MySQL schema with indexed query optimization, triggers, and DTO mappings; comprehensive JUnit 5 + Mockito unit and integration testing across all layers.

EdTech Learning Platform — React · Node.js · Express.js · MongoDB · JWT

- Full-stack RBAC platform for students, instructors, and admins; 15+ RESTful API endpoints with JWT + role guards, clean architecture, and modular folder structure.

Real-Time Chat Application — FastAPI · Next.js · WebSockets · Tailwind CSS

- WebSocket-powered FastAPI backend enabling low-latency bidirectional messaging across concurrent users; responsive Next.js frontend with real-time multi-room state management and async event handling.

ACHIEVEMENTS & HIGHLIGHTS

- LeetCode 250+ Problems Solved: Arrays, trees, graphs, dynamic programming, sliding window, and recursion — strengthening advanced problem-solving and algorithmic thinking skills.
- Design Patterns & LLD: Applied Factory, Strategy, Observer, State, Facade, Singleton patterns across academic and production projects; strong HLD and scalable architecture fundamentals.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C++

Frontend: React, Next.js, Vite, JavaFX, Tailwind CSS, Chakra UI, Zustand, React Query

Backend: Node.js, Express.js, NestJS, FastAPI, Spring Boot, RESTful API Design, Socket.IO, WebSockets

Databases & ORMs: MongoDB, PostgreSQL, MySQL, Prisma ORM, Redis

AI & NLP: Generative AI, NLP Fundamentals, Morphological Analysis, Google Gemini API, AI API Integrations

Tools & Platforms: Git, GitHub, Docker, Maven, JUnit 5, Mockito, Postman, CI/CD, VS Code

Cloud & Infrastructure: Redis, BullMQ, Stripe Integration, WebRTC

Core CS: OOP, DSA, LLD, HLD, Design Patterns, Software Engineering Principles, OS, DBMS, Problem Solving